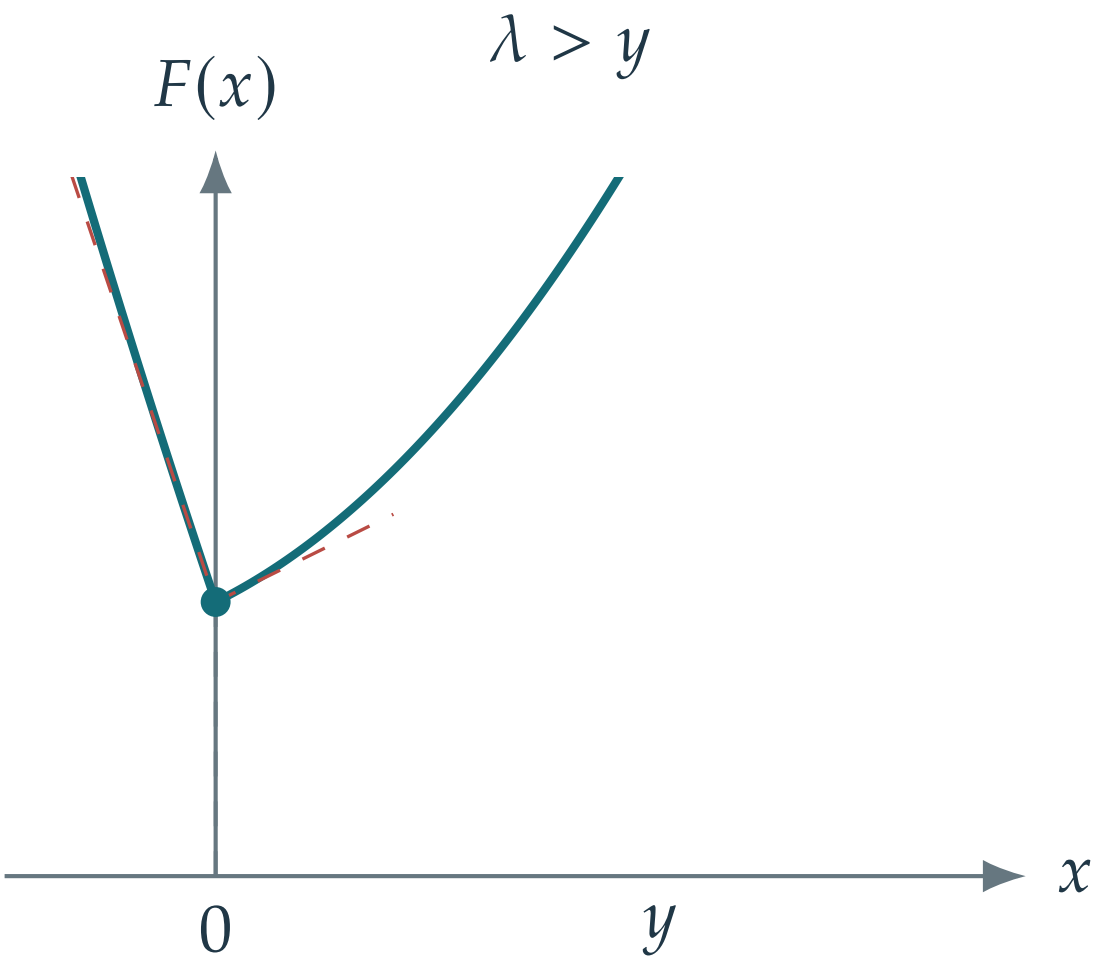

$x^\star = 0$

$y > 0$; dashed: tangents at 0